

AARHUS UNIVERSITY

DEPARTMENT OF MATHEMATICS

INTRODUCTION TO ALGEBRA 1

Handin 6

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RSA-Encryption

Let Alice have the primes $p = 107$, $q = 127$ and the exponent $k = 5$.

Determine m and $\Phi(m)$ and show that $k = 5$ is a valid choice by computing $\gcd(k, \Phi(m))$.

Determine Alice's private exponent u

A number u is a valid private exponent if it satisfies the equation $k \cdot u \equiv 1 \pmod{\Phi(m)}$.